Storyboarding

Stories user experiences in the design process

Two methods for storytelling:
1. Written scenarios
2. Storyboards

Scenario = Short, written stories about people using technology
in context with using narrative as a design tool

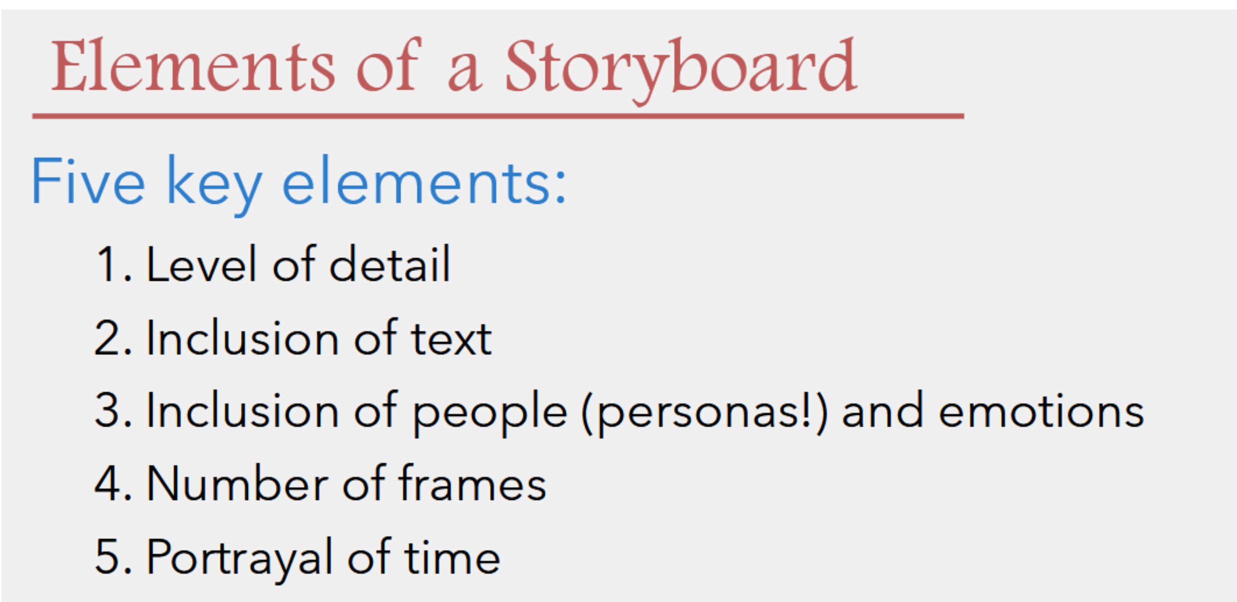
Storyboards are illustrations that represent a story.

Storyboarding is NOT the same as Paper Prototyping!

StoryBoarding Goal: Understand how your product or application fits into a larger context

Too much detail can lose universality!

Use personas!



Only include it if passage of time is a necessary part of the story!

Storyboards are a great way to bring a story to life.

Prototyping

Storyboards help generate ideas.

Prototypes help TEST those ideas.

design is an iterative cycle.

Fidelity means: How close the prototype is to the final product.

Easy access to cameras makes it easy to blur the lines between lo-fi and hi-fi prototypes

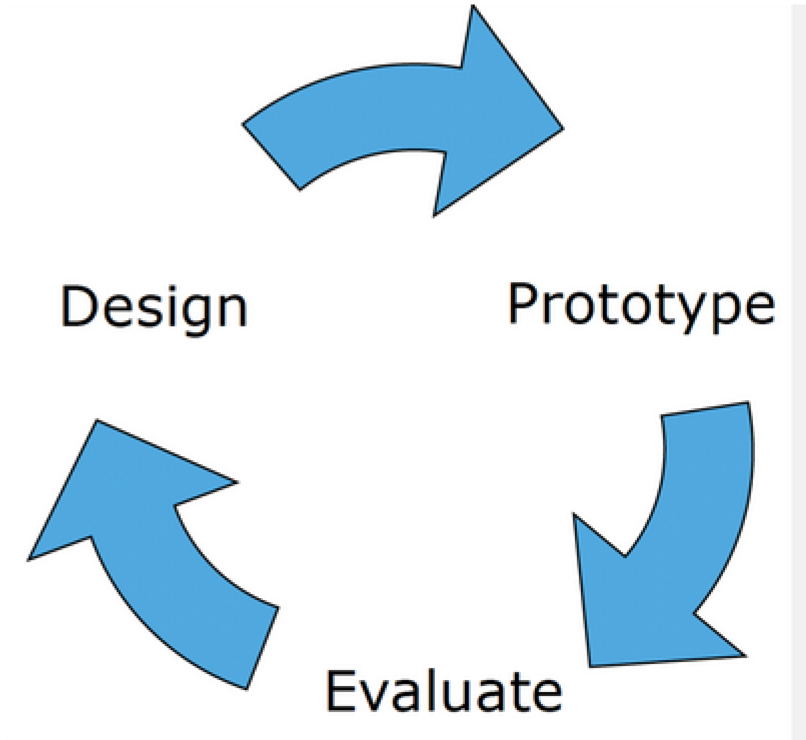
Purpose : Low fidelity's purpose explore ideas quickly

High Fidelity refines detail and validates design

Mixed Fidelity : hand-drawn screens shown on real mobile devices

Paper prototyping is a low-fidelity technique where:

- users interact with paper screens
- another person manually changes screens



Lo-fi nature forces users to consider usability issues related to layout and control

Lo fi :Focus on Content as opposed to Appearance

Observers Cannot react to user’s actions

Paper prototyping helps in :

Detects usability problems at a very early stage before implementation.

Promotes communication between stakeholders. Team members gain understanding of user needs and priorities

Hi-Fi Prototyping /Digital prototyping

design process should use multiple levels of fidelity

A very narrow prototype only represents a single feature

2. Depth

How functional and realistic the system is.

- shallow → fake responses
- deep → real logic and error handling

Depth affects:

- exploration
- user testing quality

It's generally a good idea to hold off on something that has a high fidelity look until later in the design process

3. Look

How visually realistic the prototype appears.

4. Interaction

How the prototype handles input/output. ilovepdf_merged (1)

Examples:

- clickable buttons
- animations
- gestures
- transitions

Sometimes interactions are simulated using:

- hyperlinks
- PowerPoint animations

a high fidelity prototype is a substantial time investment!

Designing for Disabled People

• 16% of US population to ages 15 to 64 is disabled.
• 10% of the workforce is disabled
• 5% of the STEM workforce is disabled
• 1% of PhDs in STEM are disabled

BD te 8-14% functional difficulties

• Up to 50% of computer users may benefit from accessibility
Features

Situational impairment : A person may not be permanently disabled but still face limitations.

“Disabled people are part of the diversity of life and are not necessarily in need of treatment or cure. They need society to remove barriers and provide access whenever possible.”

1. Prosthesis

Technology restoring lost function.

Examples:

- artificial arm
- prosthetic leg

Goal:

- “cure” or restore function

2. Assistive Technology

Technology helping people perform tasks.

Examples:

- screen magnifiers
- hearing aids
- adaptive keyboards

Focus:

- assistance

3. Access Technology

Technology enabling access regardless of method.

Examples:

- screen readers
- video phones
- wheelchairs

Focus:

- achieving goals
- enabling participation

NOT necessarily “curing.”

Disabilities Drive Innovation

• Companies are focusing on accessibility

◦ Microsoft Chief Accessibility Officer

◦ Teach Access

Challenges Faced

Memory Problems

Users may struggle to remember:

- instructions
- passwords
- navigation steps

Design Solution

Do not force users to remember too much.

Example:

- autofill
- visible instructions
- progress indicators

Attention Problems

Users may lose focus easily.

Design Solution

- simplify interfaces
- break tasks into smaller steps
- reduce clutter

Socialization Issues

Some autistic users may dislike:

- too many faces
- intense animations
- sensory overload

Design Solution

- predictable layout
- calm interfaces
- limited distractions

Challenges: Physical/mobility

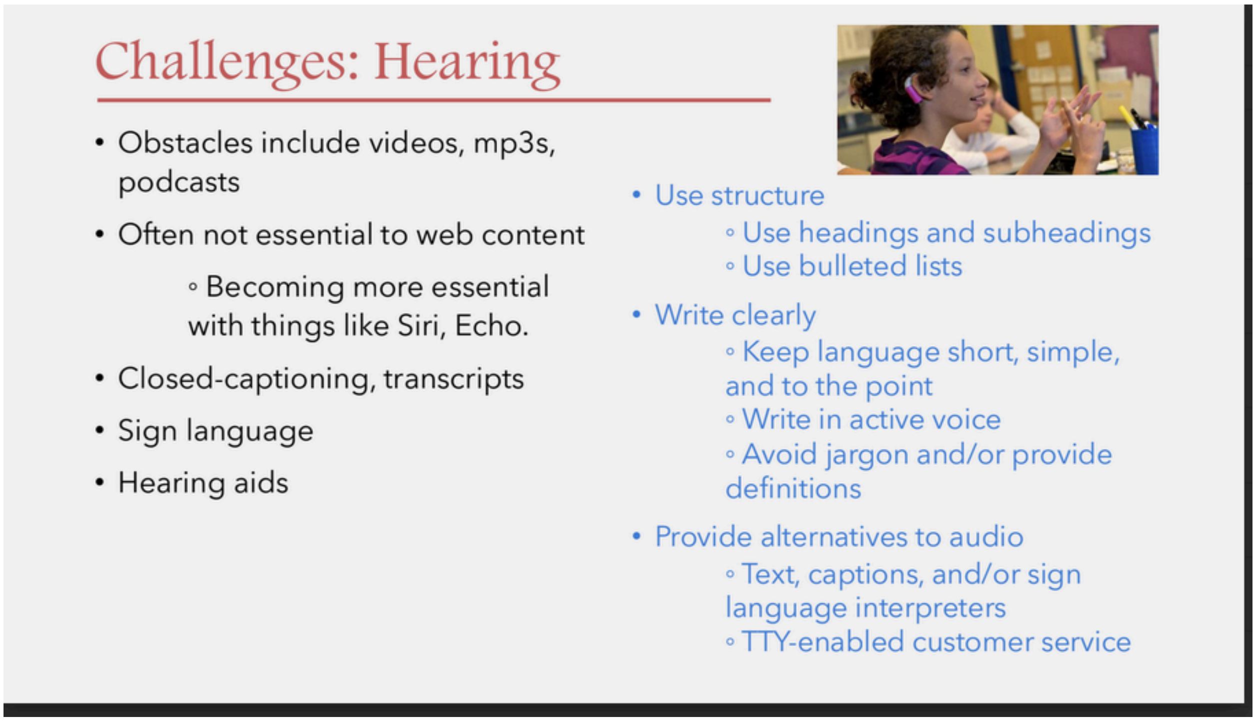
Temporary injury

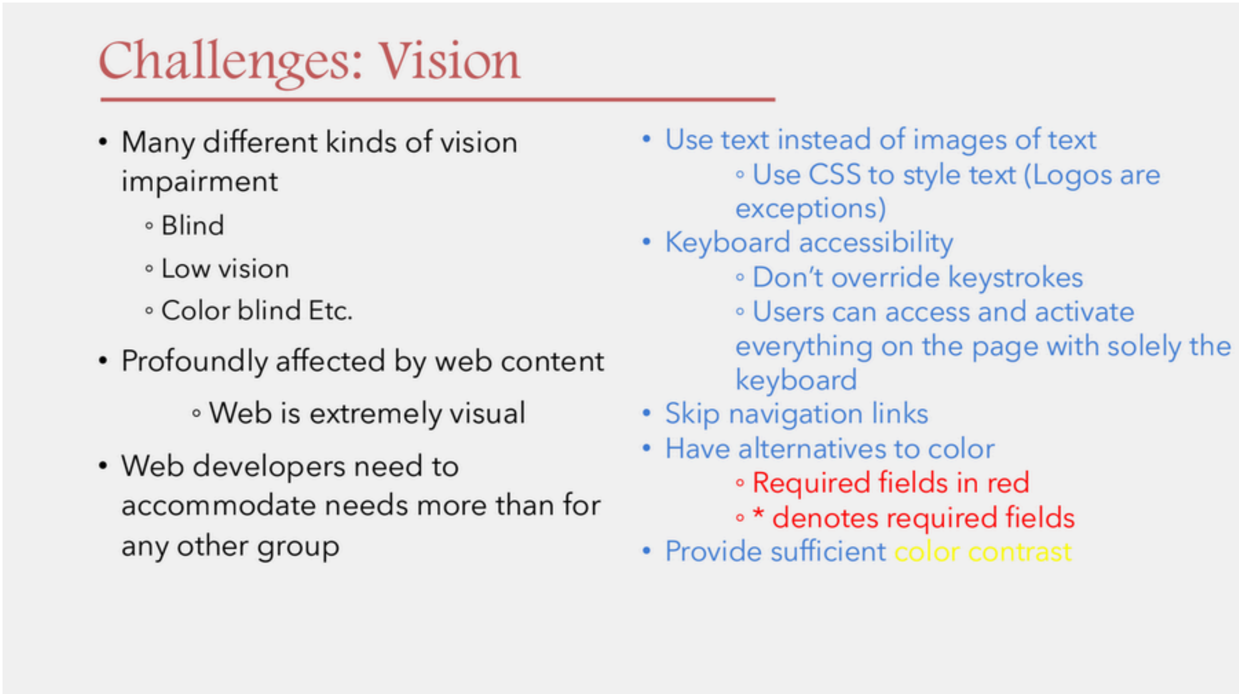
Permanent condition

1) Keyboard accessibility
2) Speech recognition compatibility
3) Provide shortcuts
4) “Skip Navigation” links

Color Blindness affects 10% male

Challenges: Hearing





Principles for Universal Design :

Many reading apps (iBooks, Kindle) enable users to personalize font size and background color. (2)

Principle 4: Perceptible Information

Information must be understandable regardless of sensory ability.

Example:

Traffic light crossing systems use:

- visual signals
- sound signals

Principle 5: Tolerance for Error

Design should reduce consequences of mistakes.

Example:

- undo button
- confirmation dialogs
- autosave

Principle 6: Low Physical Effort

System should minimize fatigue. ilovepdf_merged (1)

Example:

- sufficient spacing
- fewer repetitive actions
- easy scrolling

Principle 7: Size and Space for Use

Provide enough physical space for interaction. ilovepdf_merged (1)

Example:

- large touch targets
- wheelchair-accessible layouts

Web Accessibility

Web accessibility means:

> Designing websites and web applications so that people with disabilities can use them effectively.

The goal:

Equal access to information and interaction.

Web content is extremely visual, so people with vision impairments are particularly affected

Web developers

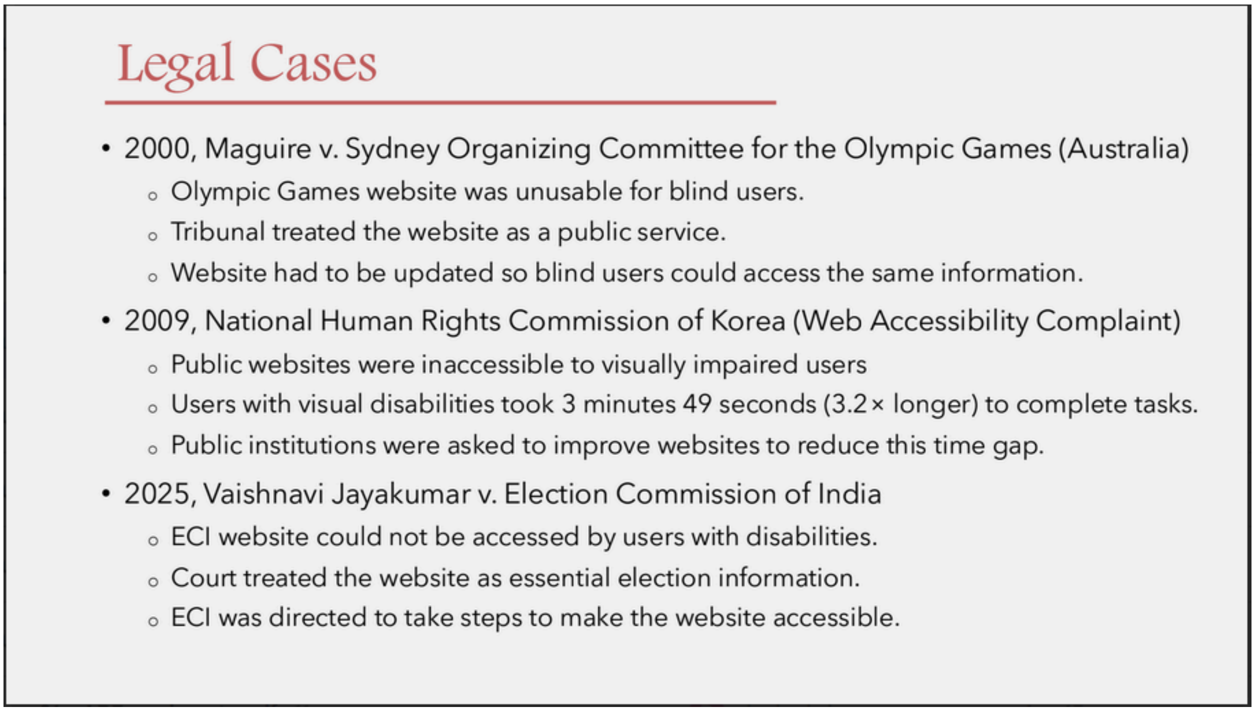
Legal Support for Accessibility

• 2009: Right to Information Act (Bangladesh)

• 2013: Rights and Protection of Persons with Disabilities Act

(Bangladesh)

• 2016: Rights of Persons with Disabilities Act (India)



P : A blind user uses a screen reader. The screen reader reads the text and describes images so the user understands the page.

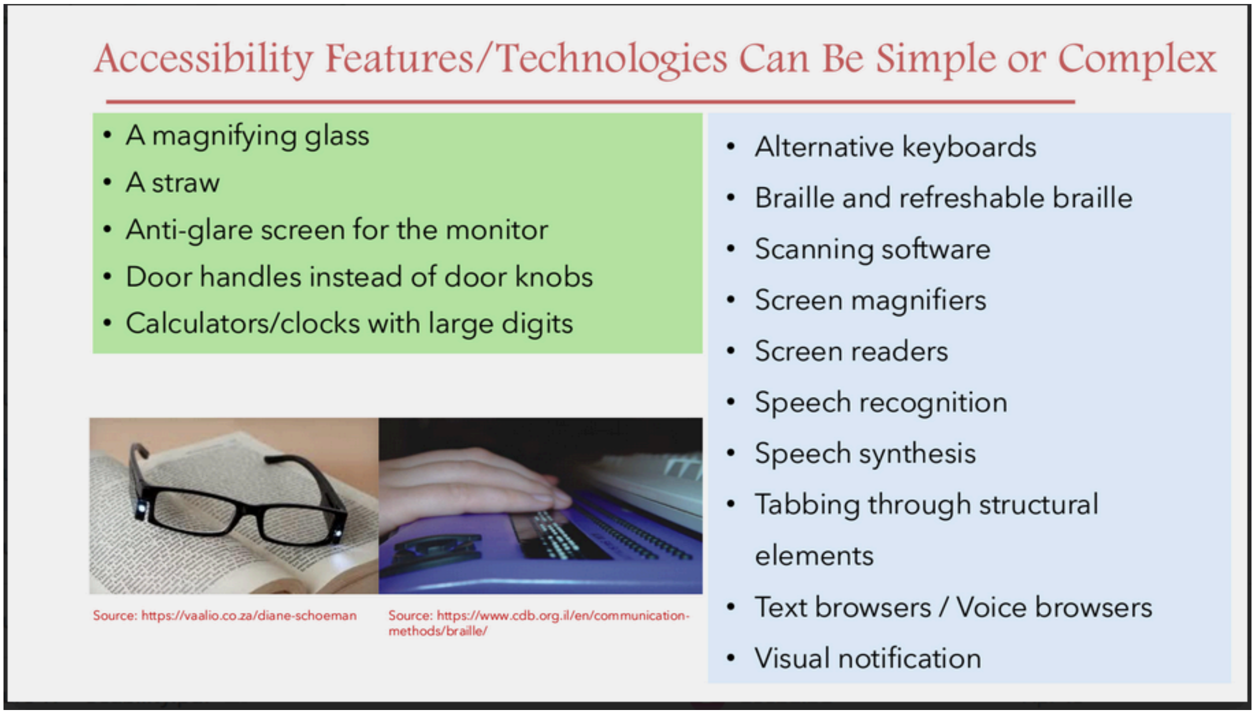
O : A person with limited hand movement uses only a keyboard to move through menus and click buttons.

U: The website layout and menu stay the same on every page.

R : A website works properly with a screen reader today and will still work when technology updates in the future.

Do NOT sacrifice accessibility for aesthetic!

Simple vs Complex



Design for marginalized communities

Do not use content that causes seizures.-----> “Little or no sustained and scaled effects on teaching, learning and achievement”

Example: Digital Green

- Problem: Teach poor farmers better farming practices
- Solution: Digital Green
  - Mediation / Mediator
  - Highly formatted, targeted video content
  - Contextual content: local presenter, not "well-dressed" scientist
  - Supporting organizations on the ground
- Outcomes: 55% adoption of new practice over 8% in old system

"Digital Green shows that technology works best when it amplifies people, institutions, and relationships that already function, rather than trying to replace them."

A Few Example Domains

- Healthcare
  - Low-cost diagnostics and telemedicine
  - Disease prevention and education
  - Healthcare informatics
- Agriculture
  - Supply chain efficiencies
  - Agricultural education
  - Market and pricing information
  - Geophysical sensing
- Education
  - Low cost computing
  - Computer sharing
  - Distance education
- Governance
  - Information organization
  - Information communication
  - Detecting and reporting corruption
  - Activism
- Design
  - Interfaces for low-literacy
  - Interfaces for low education
  - Assistive technology
- Financial services
  - Microfinance information
  - Mobile money
  - Financial literacy

context and complexities are often fundamentally different